Synthetic Difference in Differences

Critical paper talk (group meeting)

Arkhangelsky, Athey, Hirshberg, Imbens, Wager

arXiv:1812.09970 (July 2021 version)

Presenter / 2026-10-17

The aim is causal: ATT for a block-treated panel

$$\tau = \frac{1}{N_{\text{tr}}\, T_{\text{post}}} \sum_{i \in \text{tr}} \sum_{t \in \text{post}} \big(Y_{it}(1) - Y_{it}(0) \big)$$

- Unit, time, intervention: states, years, a single adoption date.
- Presenter's reading of Eq. 4.2 (block assignment). Staggered adoption is backup, not this talk.

Paper card (presenter's filling)

Field	This paper
Estimand	Block ATT (Eq. 4.2); labelled if vague
Identification	Low-rank L plus $E[E \mid W, L] = 0$
Key assumption	Assignment may depend on L, not on E
Main result	CA: SDID -15.6 (se 8.4), Table 1
Biggest threat	Placebo designs are zero-effect

Identification: knowledge vs convenience

- A1 outcome $Y = L + W \circ \tau + E$, L low rank (knowledge).
- A2 $E[E \mid W, L] = 0$ (knowledge; untestable).
- A3 unit-independence of E, homoskedasticity (convenience for placebo se).

$$E[E \mid W, L] = 0$$

What breaks

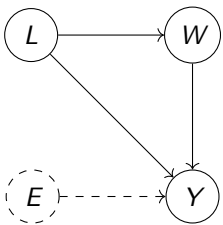
If assignment depends on E, τ is not identified; a better estimator does not close that gap.

Method: unit weights, time weights, then TWFE

$$(\hat{\tau}^{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min \sum_{i,t} \omega_i \lambda_t \big(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\big)^2$$

- ω: pre-treatment trends parallel, intercept allowed.
- λ: pre periods that predict the post mean.

DAG sketch: assignment may use L, not E



California pack-sales estimate (Table 1)

Estimator	$\hat{\tau}$	se (placebo)
SDID	-15.6	8.4
SC	-19.6	9.9
DID	-27.3	17.7

- Units: packs per person per year, 1970–2000, 38 controls.

Causal reading holds only under A1–A2

- Under A1–A2, SDID estimates the block ATT, not DID's ATT under parallel trends.
- Sensitivity: none reported for assignment-on-*E*.
- Placebo RMSE (CPS baseline): SDID 0.028, DID 0.049 (Table 2, 1000 draws). Zero-effect design.

$$\widehat{\mathrm{RMSE}}_{\mathrm{CPS}}(\mathrm{SDID}) = 0.028$$

Link to our project (fill or TODO)

- If we have few treated units and doubtful parallel trends, SDID is a robustness check beside DID and SC.
- Confirm a common adoption date before using Eq. 4.2.
- Project RQ: TODO.

$$\hat{\tau}^{\text{did}} = \arg\min \sum_{i,t} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2$$

Reproducibility: violate A2 in a placebo

$$W_i = \mathbf{1}\{E_{i1} > \text{median}(E_{\cdot 1})\}$$

- Simulate assignment that depends on E; report bias of SDID vs DID vs SC (TODO: not run in this skeleton).
- Optional 60-min segment: replicate Table 1 side by side.

What would change my mind

- A non-zero-effect simulation where SDID bias exceeds DID.
- Formal conditions for staggered adoption (appendix §8 is a recipe, not a theorem in the quickread).

$$\hat{\tau}^{\mathrm{sdid}} - \hat{\tau}^{\mathrm{did}} \quad \text{is not a test of } E[E \mid W, L] = 0.$$

Limit

Theory in the talk is block assignment only.

Discussion: should we treat SC/SDID mismatch as a warning?

- Paper: when SC and SDID disagree, SDID is usually more credible (§2.2). Quickread tag: conjectured.
- California has no ground truth.

$$\operatorname{sign}(\hat{\tau}^{\mathrm{sdid}} - \hat{\tau}^{\mathrm{sc}}) \quad \text{is not a test of A2.}$$

References

Source	Use
arXiv:1812.09970	Tables 1–2 (quickread card)
arXiv:2305.18793	Ding notation defaults

Thank you

$$\tau = E[Y(1) - Y(0) \mid \text{block assignment}]$$

Backup: staggered adoption is not this estimand

$$\hat{\tau} = \sum_{g} \pi_g \hat{\tau}_g^{\text{sdid}}$$